

Hongyi Lin

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RESEARCH PROFILE

My research centers on **world models and physical AI**. I develop interaction-driven, physics-informed models for prediction and decision-making under partial observability, with applications in autonomous driving, robotic excavation, aerial autonomy, and intelligent transportation systems.

Research interests: World Models; Physical AI; Multimodal Perception and Reasoning; Controllable Generative Modeling.

EDUCATION

Ph.D. Candidate in Mechanical Engineering 9/2022 - 6/2027 (expected)
Tsinghua University Advisor: Prof. Xiaobo Qu

- Dissertation: *Enhancing 3D Perception for Autonomous Driving in Complex Traffic Scenarios*.

- Two-time recipient of the National Scholarship for Graduate Students (top 2%).

Visiting Ph.D. Student in Transportation 8/2025 - 6/2026
Massachusetts Institute of Technology Advisors: Profs. Jinhua Zhao and Haris Koutsopoulos

B.Eng. in Automotive Engineering 9/2018 - 6/2022
South China University of Technology

- GPA: 3.89/4.00 (ranked 1st out of 111 students).

- Top 10 Outstanding Student Award, the university's highest honor for undergraduate students.

SELECTED PUBLICATIONS

Published **15 papers as first or co-first author**, most of which appeared in top-tier journals and conferences in vehicular technology, transportation, and automation.

- [1] **H. Lin**, W. Shi, H. Huang, et al. Risk-Controllable Multi-View Diffusion for Driving Scenario Generation. *CVPR VGBE Workshop*, 2026. **Best Paper Award**.
- [2] **H. Lin**, H. Chen, R. Wang, et al. A NeRF-based Road-Surface Geometry Fusion Method for Camera-only 3D Object Detection. *IEEE Transactions on Intelligent Vehicles*, 2026.
- [3] S. Zhang*, **H. Lin***, M. Wang, et al. A Dynamic Prompting and Scenario Generation Method for Autonomous Driving Perception via Large-Model Optimization. *Transportation Research Part C: Emerging Technologies*, vol. 188, Article 105672, 2026. *Equal contribution.
- [4] **H. Lin**, S. Ming, Y. Liu, et al. AEFusion: An Attention-Based Ensemble Learning Approach for BEV Fusion Perception in Autonomous Modular Buses. *IEEE Transactions on Intelligent Vehicles*, 2024.
- [5] **H. Lin**, Y. Liu, S. Li, et al. How Generative Adversarial Networks Promote the Development of Intelligent Transportation Systems: A Survey. *IEEE/CAA Journal of Automatica Sinica*, vol. 10, no. 9, pp. 1781–1796, 2023. (IF=18.3; Citations = 122)
- [6] **H. Lin**, Y. Liu, L. Wang, et al. A High-Precision Calibration and Evaluation Method Based on Binocular Cameras and LiDAR for Intelligent Vehicles. *IEEE Transactions on Vehicular Technology*, 2025.
- [7] **H. Lin**, Y. Liu, L. Wang, et al. Big Data-Driven Advancements and Future Directions in Vehicle Perception Technologies: From Autonomous Driving to Modular Buses. *IEEE Transactions on Big Data*, 2025.
- [8] **H. Lin**, C. Lyu, Y. He, et al. Enhancing State Representation in Multi-Agent Reinforcement Learning for Platoon-Following Models. *IEEE Transactions on Vehicular Technology*, vol. 73, no. 8, pp. 12110–12114, 2024.

- [9] H. Lin, Y. He, Y. Liu, et al. Deep Demand Prediction: An Enhanced Conformer Model with Cold-Start Adaptation for Origin-Destination Ride-Hailing Demand Prediction. *IEEE Intelligent Transportation Systems Magazine*, vol. 16, no. 3, pp. 111–124, 2024.
- [10] P. Liu, H. Lin, Y. Zhao, et al. DesEAD: Enhancing End-to-End Autonomous Driving with Scene Descriptions. *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 2025.
- [11] X. Qu, H. Lin, Y. Liu. Envisioning the Future of Transportation: Inspiration of ChatGPT and Large Models. *Communications in Transportation Research*, vol. 3, 2023. (Citations = 143)

Papers Under Review

- [M1] H. Lin, Y. Liu, J. Zhao, et al. Reframing Vision-Language Models as Bounded Optimizers Rather than Direct Regressors in Autonomous Driving. Submitted to *Nature Mechanical Engineering*; under review.
Earlier version: *Rethinking Foundation Model Collaboration: Enhancing Specialized Models through Proxy Task Reasoning*, arXiv:2606.31157, 2026.
- [M2] H. Lin, H. Chen, J. Wang, et al. AiONet: A Unified Vehicle-Infrastructure Cooperative Perception Framework for Urban Mobility Systems. Submitted to *Transportation Research Part C: Emerging Technologies*; minor revision.
- [M3] H. Lin, S. Zhang, Y. Liu, et al. PileBelief: Persistent Physical State for Interaction-Driven World Modeling. Submitted to *International Conference on Learning Representations (ICLR 2027)*.
- [M4] H. Lin, Y. Liu, Q. Kang, et al. VeriFuse: Bounded Vision-Language Arbitration and Reason-Guided Refinement for Cooperative 3D Perception. Submitted to *IEEE International Conference on Robotics and Automation (ICRA 2027)*.
- [M5] H. Lin, S. Zhang, X. Liu, et al. Physical-Touch Observability from Wrist Wrench in Granular Scooping. Submitted to *IEEE International Conference on Robotics and Automation (ICRA 2027)*.

SELECTED RESEARCH PROJECTS

World Models for Sustained Autonomous Operation of Construction Machinery

Primary contributor

6/2026 - Present

Partners: SAIC Motor Corporation; Tsing AI (Shanghai)

- Led the development of PileBelief, a world model for autonomous excavation under partial observability.
- Used interaction history to predict terrain changes, payload, and contact forces for action selection; validated the system in consecutive real-world excavation trials.
- Reduced joint prediction error by 71.8% and improved top-1 action-selection accuracy from 26.25% to 62.50%.

Cognitive Enhancement Theory and Progressive Learning Models for Autonomous Driving

Primary student contributor

9/2025 - Present

Ministry of Education Pioneer Project; subproject funding: RMB 6.26 million.

- Developed a risk-controllable driving scenario generator, achieving a scenario duplication rate of $\leq 25\%$ and an 80% reduction in the cost of obtaining valid safety-critical samples.
- Developed a unified framework for multimodal cooperative perception across multiple sensing platforms.

VLA-Enabled RoboBus Deployment for Smart Public Mobility

Primary student contributor

6/2025 - Present

Hangzhou Airport Economic Zone; funding: RMB 50 million.

- Designed VLA architectures for RoboBus scene understanding, decision support, and driving interactions.
- Built a dataset covering a 10.7 km route, supporting generation of scenarios involving bus arrivals and departures at stops, encounters on narrow roads, and pedestrian-vehicle interactions.
- Compiled two years of accident data from 12 cities to support generation of long-tail hazardous scenarios and extreme interactions.

Modular and Autonomous Buses for Sustainable Urban Transportation

Primary student contributor

1/2023 - Present

NSFC Key International (Regional) Joint Research Program; funding: RMB 2.60 million.

- Designed a mechanical linkage for in-motion docking between two autonomous modular buses.
- Developed a 3D perception method incorporating road-geometry priors.
- Proposed a cold-start travel-demand prediction method for areas without historical demand data.

Control of Advanced Driver Assistance Systems for Intelligent Vehicles

Project leader

5/2020 - 5/2021

National Undergraduate Innovation Training Project, Ministry of Education, China.

- Built an autonomous-driving experimental platform integrating localization, perception, decision-making, and control, with validation in simulation and real-vehicle experiments.
- Implemented a neural-network-based trajectory planner and deployed it on a physical vehicle.

INDUSTRY RESEARCH EXPERIENCE

Suzhou Automotive Research Institute, Tsinghua University

Research Intern, Autonomous Driving Perception

6/2024 - 8/2024

- Improved the threshold-constraint module in RenderOCC and developed a reusable visualization module for 3D occupancy prediction.
- Developed an automatic annotation pipeline combining a VLM-based selector with specialized models, improving 3D annotation AP by 33%, reducing the manual review rate to 9%, and saving approximately RMB 500,000 in annotation costs.

Tsing AI (Shanghai) Co., Ltd.

Model Engineer, Perception Team

2/2023 - 8/2023

- Designed a staged docking strategy and a LiDAR-camera BEV perception framework for in-motion docking of autonomous modular buses.
- Collected 18,120 multimodal frames across 29 scenario categories using a 1:2-scale dual-vehicle platform, covering inter-vehicle distances of 0.5–50 m and diverse weather and road conditions.
- Achieved longitudinal position and heading MAEs of 2.3 cm and 0.85° on the collected test set.

TEACHING AND MENTORING

MIT-UF-NEU 2026 Joint Summer Research Camp

Research Mentor

Summer 2026

- Mentored two students on research projects that resulted in two manuscripts submitted to ICRA.

Future Transport Modes, Tsinghua University

Teaching Assistant

Fall 2023 & Fall 2024

Instructors: Prof. Xiaobo Qu and Prof. Kai Wang

- Assisted with course design and organization, project mentoring, and assignment evaluation.

Selected Students Mentored

- **Graduate students:** Hanyang Tang (MIT, 2025); Ziyu Lin (Tsinghua University, 2026); Ruo Chen (Zhejiang University, 2025); Shouqun Ming (Chongqing University → UBTECH Robotics).
- **Undergraduate students:** Yiyao Liu (UESTC, 2026); Qunling Han, Hanyi Qiu, and Danqi Zhao (Tsinghua University, 2024 → Ph.D. students at Tsinghua University).

ACADEMIC SERVICE

- Session Chair, Smart Transportation and Vehicle Equipment Symposium, Hangzhou, 2024.
- Contributed to editorial and outreach activities for *COMMTR* and *JICV*, including manuscript solicitation, peer review, special-issue planning, and international journal promotion.
- **Journal reviewer:** TR Part-A, TR Part-C, TR Part-E, IEEE T-IV, IEEE T-ASE, IEEE T-MC, IEEE T-VT, IEEE ITSM, IET ITS, *COMMTR*, *JICV*.
- **Conference reviewer:** ICML, IEEE IV, IEEE ITSC, WCTR, ITS WC, APTE.

AWARDS AND HONORS

- Best Paper Award, CVPR VGBE Workshop 2026
- First Prize, Autonomous Driving Track, Tsinghua University AI Competition (1st/80) 2025
- National Scholarship for Graduate Students, Ministry of Education of China 2024, 2025
- First Prize, Robot Application Track, China Robotics and AI Competition 2024
- Champion, Real Car Invitational Competition, China Robotics and AI Competition 2024
- Outstanding Oral Presentation, Doctoral Forum, Tsinghua University 2024
- Comprehensive First-Class Scholarship, Tsinghua University 2023
- Finalist, COMAP Interdisciplinary Contest in Modeling (ICM)(top 1%) 2021
- Top 10 Outstanding Student Award, South China University of Technology (2nd/20,000+) 2020
- National Scholarship for Undergraduate Students, Ministry of Education of China 2019, 2020

TALKS AND PRESENTATIONS

- “Risk-Controllable Multi-View Diffusion for Driving Scenario Generation,” CVPR VGBE Workshop, Denver, USA, 2026.
- “Integrating Physics into Scenario Generation for Autonomous Driving and UAVs,” KAIST, South Korea (online), 2026.
- “VLMs as Selectors Rather than Regressors in Autonomous Systems,” MIT SRB Forum, Cambridge, USA, 2026.
- “Bidirectional Physical Field Generation for Low-Altitude-Aware Urban Planning,” MIT JTL Lab Seminar, Cambridge, USA, 2025.
- “Enhancing Transportation Modeling with Physics-Informed Machine Learning,” APTE, NUS, Singapore, 2024.
- “Framework for Large-Scale Urban Traffic State Estimation Based on AIGC,” KES-STs, Portugal (online), 2024.
- “Enhancing State Representation in Multi-Agent Reinforcement Learning for Platoon-Following Models,” WTC, Qingdao, China, 2024.

SELECTED PATENTS

- [P1] X. Qu, **H. Lin**, Y. Liu, et al. A LiDAR-based Perception Method and Devices for Multi-Vehicle Platooning in Intelligent Connected Buses. Chinese Patent, ZL202410433936.7.
- [P2] X. Qu, **H. Lin**, Y. Liu, et al. A Camera-based Perception Method and Devices for Multi-Vehicle Platooning in Intelligent Connected Buses. Chinese Patent, ZL202410433937.1.
- [P3] X. Qu, **H. Lin**, Y. Liu, et al. A Control Method and Devices for Platooning Docking of Intelligent Connected Buses. Chinese Patent, ZL202410410294.9.
- [P4] K. Wang, X. Qu, **H. Lin**. A Vehicle Speed Control Method, Apparatus, Device, and Storage Medium. Chinese Patent, ZL202310639458.0.
- [P5] W. Zhang, M. Xu, **H. Lin**, et al. A Connection Device for Modular Cars. Chinese Patent, ZL202420273708.3.

REFERENCES

Prof. Xiaobo Qu xiaobo@tsinghua.edu.cn
Member, Academia Europaea – The Academy of Europe;
Chair Professor of Intelligent Transportation, School of Vehicle and Mobility, Tsinghua University.

Prof. Jinhua Zhao jinhua@mit.edu
Professor of Cities and Transportation; Head, Department of Urban Studies and Planning;
Founder, MIT Mobility Initiative, Massachusetts Institute of Technology.